

HEALTHCONNECT EXPERIENCE LAB

Project Title: Healthconnect Clinic Project Summary
Program: AnalystLab Africa Data Analytics Internship
Week 4

Project Overview

HealthConnect Clinic is an appointment-based healthcare provider seeking to reduce missed appointments and improve the patient support experience.

The central project question is: How can HealthConnect Clinic use data to reduce missed appointments and improve the patient support experience?

As a Data Analytics intern, my Week 4 responsibility is to understand the appointment dataset, assess its quality, identify variables relevant to appointment attendance and define an initial analytical approach.

Specific Problem Addressed

The Data Analytics track will investigate appointment attendance and no-show patterns. The analysis will seek to determine which demographic, appointment, booking, reminder and historical behaviour factors are associated with missed appointments. The ultimate objective is to provide an evidence-based foundation for improving appointment management and patient engagement.

Resources Used

The following project resources were reviewed:

- HealthConnect Appointment Dataset
- HealthConnect Data Dictionary

The original project resources will be preserved, while any cleaned or transformed datasets will be maintained separately.

Key Week 4 Observations

Initial inspection established that the dataset contains:

- 5,000 appointment records
- 18 variables

The dataset provides information covering:

- Patient demographics
- Appointment details
- Booking behaviour
- Previous appointment history
- Reminder activity
- Distance to clinic
- Waiting time
- Appointment outcomes

Initial missing-value checks identified:

- 90 missing Distance to Clinic records (1.8 %)
- 60 missing Waiting Time records (1.2%)

Proposed Analytical Approach

The proposed analysis will proceed through:

- Data validation
- Data cleaning and preparation
- Exploratory data analysis
- KPI development
- Visual analysis
- Business interpretation
- Recommendations

The analysis will focus particularly on appointment outcome, previous no-shows, reminder activity, booking lead time, appointment type, appointment timing, demographic segments and operational variables.

Proposed KPIs

The following KPIs have been identified for subsequent stages:

- No-Show Rate
- Attendance Rate
- Cancellation Rate
- Reminder Coverage Rate

These KPIs were selected because they directly relate to the project's central objective of understanding and reducing missed appointments.

Key Considerations

Several factors may affect the analysis:

- Missing values in Distance to Clinic and Waiting Time.
- Inconsistencies in date formatting.
- Need to validate appointment ID uniqueness.
- Potential inconsistencies in categorical values.
- Limited contextual information explaining why patients miss appointments.
- The fictional and anonymised nature of the dataset.
- The distinction between genuine missing values and meaningful categories such as None in Reminder Channel.

Week 5 Proposed Focus

The proposed Week 5 focus is data preparation and exploratory analysis.

Activities will include:

- Completing data-quality validation.
- Standardising date fields.
- Reviewing and resolving genuine duplicate records if identified.
- Assessing categorical consistency.
- Determining appropriate treatment for missing values.
- Exploring appointment outcomes across key variables.
- Beginning KPI calculations.
- Identifying initial no-show patterns.

Conclusion

Week 4 established the analytical foundation for the HealthConnect Experience Lab.

The initial review confirms that the available dataset contains relevant information for investigating appointment attendance and no-show behaviour. Initial data-quality checks have identified limited missing values, while further validation is required for duplicates, date consistency and other potential quality issues.

The next stage will move from understanding and planning into data preparation and exploratory analysis, with the goal of identifying actionable patterns that can eventually support HealthConnect Clinic in reducing missed appointments and improving patient support.